

- Fig. 8.1 is a diagram of part of a guard cell showing some of the events that occur when ABA binds to its receptor.

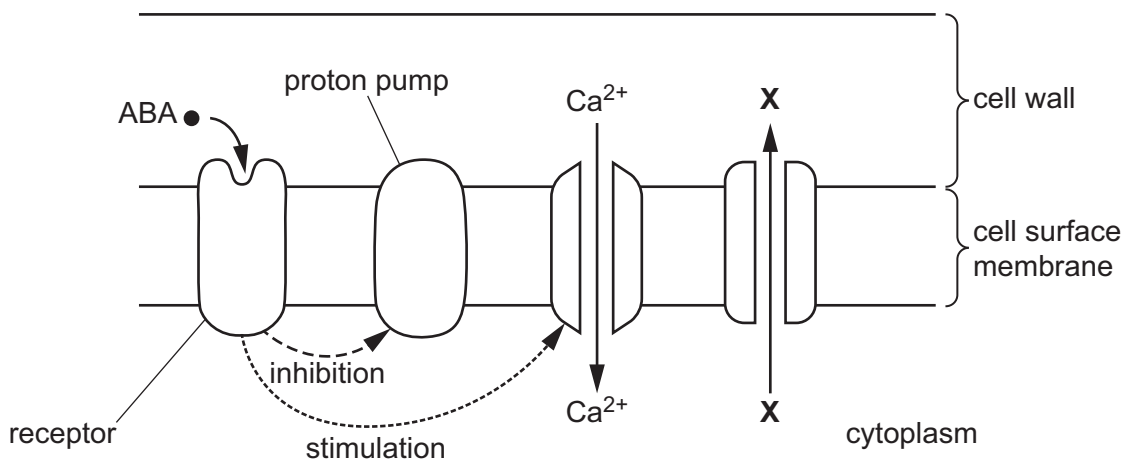


Fig. 8.1

Fig. 8.1 shows that calcium ions are involved in the events within a guard cell that result in stomatal closure.

Outline the role of calcium ions in the response of the guard cell to a reduced water supply **and** explain how this response results in stomatal closure.

Identify **X** in your answer.

[4]



- (b) Explain, using examples, why the homeostatic control involving the opening and closing of stomata is important for the efficient functioning of plants.

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[Total: 8]